Jaewook J. Suh

CONTACT INFORMATION

Department of Applied Mathematics & Statistics
Johns Hopkins University

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RESEARCH INTERESTS

AI- and Computer-Assisted Algorithm Analysis, Convex Optimization, Monotone Operator Theory, Adaptive First-Order Methods, Accelerated Algorithms, Optimization for Machine Learning

ACADEMIC APPOINTMENTS & AFFILIATIONS

Johns Hopkins University 2026 – Present
Lecturer, Department of Applied Mathematics & Statistics

Rice University 2024 – 2026
Postdoctoral Researcher
- Worked with Professor Shiqian Ma `[website]`

EDUCATION

Seoul National University 2024
Ph.D. in Mathematical Sciences
- Advisor: Professor Ernest K. Ryu `[website]`

Seoul National University 2018
B.S. in Mathematics and Physics
Teacher Certification in Mathematics and Physics

PUBLICATIONS

Published

1. **J. J. Suh**[*], T. Yoon[*], E. D. H. Nguyen, B. Ying, S. Ma. *Peppy: An* AI*-Assisted Workflow for Tight Convergence Analysis of Optimization Algorithms.* ICML Workshop: AI as a Tool for Mathematics, Computer Science, and Machine Learning, 2026. *Workshop version.* [*]Lead authors
2. R. Park, **J. J. Suh**, Y. Hong, E. K. Ryu. *Numerical Analysis of HiPPO-LegS ODE for Deep State Space Models.* Transactions on Machine Learning Research, 2026.
3. J. Park[*], **J. J. Suh**[*], B. Wang, A. Bhattacharya, S. Ma. *Adaptive Gradient Descent on Riemannian Manifolds and Its Applications to Gaussian Variational Inference.* ICLR, 2026. [*]lead authors
4. **J. J. Suh**, B. Ying, X. Jiang, E. D. H. Nguyen. *PEPFlow: A Python Library for the Workflow of Performance Estimation of Optimization Algorithms.* NeurIPS Workshop on GPU-accelerated and Scalable Optimization, 2025. `[Website]` `[GitHub]`
5. S. P. Boyd, T. Parshakova[*], E. K. Ryu, and **J. J. Suh**[*]. *Optimization Algorithm Design via Electric Circuits.* NeurIPS (spotlight, 2.1%), 2024. `[GitHub]` [*]Lead authors (alphabetical)
6. T. Yoon, J. Kim, **J. J. Suh**, and E. K. Ryu. *Optimal Acceleration for Minimax and Fixed-Point Problems is Not Unique.* ICML (spotlight, 3.5%), 2024.
7. **J. J. Suh**, J. Park, and E. K. Ryu. *Continuous-time Analysis of Anchor Acceleration.* NeurIPS, 2023.
8. **J. J. Suh**, G. Roh, and E. K. Ryu. *Continuous-time Analysis of Accelerated Gradient Methods via Conservation Laws in Dilated Coordinate Systems.* ICML (long presentation, top 2%), 2022.

Preprints

1. **J. J. Suh**, T. Yoon, E. D. H. Nguyen, B. Ying, S. Ma. *Peppy: An* AI*-Assisted Workflow for Tight Convergence Analysis of Optimization Algorithms.* arXiv preprint (full version), 2026.
2. T. Yoon[*], **J. J. Suh**[*], E. D. H. Nguyen, B. Ying, S. Ma. *Toward a Systematic Understanding and Interactive Search of Lyapunov-Style Proofs in Optimization.* 2026. [*]Lead authors

3. E. D. H. Nguyen, **J. J. Suh**, X. Jiang, S. Ma. *Exact worst-case convergence rates for Douglas–Rachford and Davis–Yin splitting methods.* 2025.
4. **J. J. Suh**, S. Ma. *An Adaptive and Parameter-Free Nesterov's Accelerated Gradient Method for Convex Optimization.* 2025.

Research Software & Tools

Peppy
- AI-assisted workflow for convergence analysis of optimization algorithms, built on PEPFlow

PEPFlow `[Website]` `[GitHub]` 2025 – Present
Co-developer
- Open-source Python library for computer-assisted analysis of optimization algorithms
- Provides infrastructure for interactively constructing convergence proofs via the PEP workflow
- Supports symbolic and numerical verification of worst-case convergence guarantees

Invited Talks & Seminars

Peppy: An AI-Assisted Workflow for Tight Convergence Analysis of Optimization Algorithms
- KAIST Graduate School of AI for Math Seminar — Jul 2026

Adaptive and Tuning-free First-Order Methods via Lyapunov Analysis
- Institute of Artificial Intelligence, University of Central Florida — May 2026
- Department of Combinatorics & Optimization, University of Waterloo — Feb 2026

Optimization Algorithm Design via Electric Circuits
- Quantitative Methods Seminar, Purdue University — Mar 2025
- KIAS Center for AI and Natural Sciences Seminar — Jan 2025

Teaching Experience

Lecturer, Johns Hopkins University
- Introduction to Convexity (EN.553.665) — Fall 2026

Substitute Lecturer, Rice University
- Iterative Methods for Systems of Equations and Unconstrained Optimization — Spring 2025

Instructor, Seoul National University
- Exercises in Introduction to Mathematical Analysis I — Spring 2020
- Exercises in Introduction to Mathematical Analysis II — Fall 2020
- Exercises in Calculus I — Spring 2019, Spring 2021
- Exercises in Calculus II — Fall 2018, Fall 2019, Spring 2022
  - Average student course evaluation score: 9.96/10

Conference Presentations

Peppy: An AI-Assisted Workflow for Tight Convergence Analysis of Optimization Algorithms
- ICML Workshop AI4Research (Poster) — Jul 2026

Adaptive Gradient Descent on Riemannian Manifolds and Its Applications to Gaussian Variational Inference
- International Conference on Learning Representations (Poster) — Apr 2026

PEPFlow: A Python Library for the Workflow of Performance Estimation of Optimization Algorithms
- SIAM Conference on Optimization — Jun 2026
- Cornell ORIE Young Researchers Workshop (Poster) — Oct 2025

An Adaptive and Parameter-Free Nesterov's Accelerated Gradient Method for Convex Optimization

- INFORMS Optimization Society Conference (IOS) Mar 2026
- INFORMS Annual Meeting Oct 2025
- International Conference on Continuous Optimization (ICCOPT) Jul 2025

Optimization Algorithm Design via Electric Circuits

- AI-Owls meeting (Rice University) Mar 2025
- Neural Information Processing Systems (Poster) Dec 2024

Optimization Algorithm Design by Continuous-time Analysis

- KMS Spring Meeting Apr 2024
- UNIST Workshop Optimization & Machine Learning Aug 2023

Continuous-time Analysis of Anchor Acceleration

- Neural Information Processing Systems (Poster) Dec 2023
- CJ CheilJedang Tech Talk Jul 2023

Continuous-time Analysis of AGM via Conservation Laws in Dilated Coordinate Systems

- SIAM Conference on Optimization May 2023
- Samsung AI Forum (Poster) Nov 2022
- INFORMS Annual Meeting Oct 2022
- International Conference on Machine Learning Jul 2022

Professional Service

Conference Organization

- Minisymposium Organizer, SIAM Conference on Optimization (SIOPT), 2026
  - "AI- and Computer-Assisted Optimization"
- Session Organizer, INFORMS Optimization Society Conference (IOS) 2026
  - "Systematic Analysis of Optimization Algorithms Leveraging Computer-Assisted Proofs"

Reviewer

- ICML 2026 Workshop: AI as a Tool for Mathematics, Computer Science, and Machine Learning
- Journal of Machine Learning Research (JMLR)
- Journal of Computational and Graphical Statistics (JCGS)
- Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- Mathematical Programming
- Journal of Optimization Theory and Applications
- Journal of Scientific Computing
- NeurIPS Workshop on GPU-accelerated and Scalable Optimization (ScaleOPT)

Awards & Honors

Postdoctoral Fellowship Program (Nurturing Next-Generation Researchers), granted by the National Research Foundation of Korea (NRF) 2024

Google Travel Grant Award to attend ICML 2022

Excellent Teaching Assistant Award 2019

National Scholarship for Science and Engineering 2013

Additional Teaching Assistant Experience

Teaching Assistant, Seoul National University

- Mathematical Foundations of Deep Neural Networks Spring 2024
- Calculus for Life Science I Fall 2023
- Differential Equations Spring 2023
- Mathematical and Numerical Optimization Fall 2021, Fall 2022
- Linear Algebra Spring 2021
- Mathematics for Economics and Business Spring 2020, Fall 2020
- Introduction to Mathematical Analysis I, II Spring 2020, Fall 2020
- Engineering Mathematics I, II Summer 2019, Fall 2019

- Calculus I, II — Fall 2018–Fall 2019, Spring 2021, Spring 2022

Teaching Practicum, Seoul National University Middle School (Physics) — May 2016
- Completed a one-month student teaching experience at a middle school
- Gained hands-on experience in classroom instruction and student engagement

Mentorship & Outreach

Undergraduate Research Mentorship
- Collaborated with undergraduate interns on research (published at ICML 2022, 2024)

Seoul National University College of Natural Science Camp (TA, Mathematics) — Aug 2019
Intercollegiate Education Outreach Club Activity — 2012–2014
- Visited various middle and high schools across the region to mentor students
- Provided studying tips on an online website through columns and Q&A
- Served as Club Officer (Fall 2013) and President (Spring 2014)